Patrick Salsbury

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EXPERIENCE

Data Engineer | *Tesla* - Palo Alto, CA — January 2025 - Present

- Lead the proof-of-concept, design, and development of a **ClickHouse** database for time-series data generated from 100+ hardware devices, achieving 20x faster analytical queries compared to **Iceberg**; this enabled engineers to perform data analysis faster, increasing productivity.
- Deployed a REST API using **FastAPI** with Pydantic to enable 100+ devices to store test metadata and metrics efficiently.
- Designed and implemented a **MySQL** database to model the logging, tracking, and data analysis of vehicle part components for a wide range of reliability tests the Hardware Test team conducts.
- Built and maintained scalable data pipelines using **Airflow** and **Spark** to process petabytes of vehicle data and terabytes of hardware test data, integrating **MySQL**, **S3** storage, **Iceberg** datasets, and external APIs to fulfill requests for various stakeholders.

Data Science Intern | *Tesla* - Palo Alto, CA — June 2024 - December 2024

- Developed a comprehensive internal data analytics application for the hardware test team, enabling them to monitor device usage, analyze key performance indicators, and get notified of statistically significant anomaly data points during live tests; using **Plotly Dash** hosted on a **Linux** server.
- Analyzed over 10 billion entries of fleet data to identify root causes of vehicle fault errors; went forward and created an interactive dashboard using **Bokeh Panel** and **Trino** to showcase these different key performance indicators and allow users to identify issues themselves.
- Collaborated with test engineers and project managers from various teams to understand and meet various data goals ranging from ad-hoc analysis reports to entire new dashboard creations.

Data Science & Engineering Intern | *EverCharge* - Palo Alto, CA — June 2023 - September 2023

- Identified bugs, hardware issues, and installation problems that were previously unknown affecting 5-10% of all products by performing time series analysis, inferential analysis, and anomaly detection using **Python** and **SQL**.
- Engineered **Retool** dashboards to showcase **KPIs** regarding device performance while performing **PostgreSQL** query optimization.
- Published 3 different data science reports that communicated answers to stakeholders' questions and highlighted areas of concern.
- Initiated a proof-of-concept project that involved training an ML anomaly detection model to identify abnormal device behavior.

Data Science Intern | *HM Electronics* - Carlsbad, CA — June 2022 - September 2022

- Analyzed customers' device telemetry data, improving production quality and customer relationships by showcasing areas of concern.
- Extracted semi-structured console log data from 100+ customer devices to ingest into a **Azure Blob Data Lake**.
- Automated the **ETL** process on 30+ million raw records of data using **PySpark** on **Databricks(Microsoft Azure)**.
- Predicted customer product device failure using **Classification** and **Clustering ML** models, isolating variable indicators for tech support.
- Led **Agile** sprint presentations to communicate findings using **Python**, **SQL**, and **Power BI** to influence decision-making of product managers and software engineers.

PROJECTS

Mitigating Gender Bias in CHD Prediction - *Capstone Project*

- Designed and implemented a feed-forward neural network to predict Coronary Heart Disease using CDC's NHANES dataset with over 37,000 records.
- Integrated an adversarial debiasing framework to mitigate gender bias by penalizing correlations between predictions and sensitive attributes.
- Improved balanced accuracy scores from 0.71 to 0.77 and reduced equal opportunity difference by 95.6%.

E-commerce Recommender System

- Developed a machine learning model for an e-commerce recommender system using **Sci-kit Learn**, **H2o AutoML**, and **Tensorflow.**
- Performed exploratory data analysis, feature engineering, model training and validating, and hyperparameter tuning in order to obtain a model with an accuracy, F1-score, and ROC-AUC of 0.71, 0.72, 0.79 respectively.
- Deployed the finalized Collaborative Filtering Tensorflow model using **Flask** and **HTML** to emulate a real-world recommender system user experience.

Real-Time Weather Data Dashboard Pipeline

- Constructed dynamic **dashboards** visualizing real-time weather updates using **Tableau**, **MongoDB, Docker,** and weather APIs.
- Highlighted temperature, humidity, and other attributes using **Apache Airflow** to build an ETL **batch** processing pipeline that ingested, transformed, and loaded the data while performing various quality checks.

Popular Youtube Video Titles Generator

- Utilized Youtube APIs to store and analyze daily top 50 trending videos to generate new possible trending video titles.
- Established an automated **batch ETL pipeline** to populate an **SQLite database** using **Bash Scripting, SQL,** and **Python**.
- Generated titles using sentence linguistics, **unigram NLP models**, and **N-gram NLP models**.

EDUCATION

University of California, San Diego | B.S. Data Science — September 2021 - June 2024
De Anza College | A.S. Computer Science — September 2018 - June 2021

TECHNICAL SKILLS

Languages: Python, SQL, Java
Databases: Iceberg, ClickHouse, MySQL PostgreSQL
Libraries / Frameworks: Pandas, PySpark, SQLAlchemy, SciKit-Learn, SciPy, Dash, Plotly, XGBoost, TensorFlow, PyTorch
Softwares: Git, Docker, Databricks, Azure, Power BI, Tableau, Retool, Apache Spark, Apache Airflow